

Evolutionary Computation and its Applications

Today's goals



Module 1

Fundamental Concepts in Optimization

Module 2

Genetic Algorithm

Module 5 Introduction

✔

Introduction to Module 5's Study

Video • 1 min

✔

How to Access MATLAB Online

Reading • 5 min

Module 5-1 Genetic Algorithm 1

✔

Genetic Algorithm: A First Introduction

Video • 6 min

✔

Genetic Algorithm: A Demo

Video • 8 min

✔

Materials on Genetic Algorithms

Reading • 10 min

✔

How Genetic Algorithms Solve Optimization Problems

Reading • 5 min

Module 5-2 Genetic Algorithm 2

✔

Genetic Algorithms: GA Representation, Initialization and Selection

Video • 10 min

✔

Genetic Algorithms: GA Crossover and Mutation

Video • 7 min

✔

Genetic Algorithms: GA Stopping Criterion and Summary

Video • 5 min

✔

Materials on Genetic Algorithms

Reading • 10 min

✔

How GA Operators Work Together

Reading • 5 min

Module 5-3 Use Genetic Algorithm 1

●

Use GA via MATLAB: GA Function

Video • 7 min

●

Use GA via MATLAB: Employ the Function

Video • 6 min

●

Materials on Using Genetic Algorithms

Reading • 10 min

●

Using Genetic Algorithms Effectively in MATLAB

Reading • 5 min

Module 5-4 Use Genetic Algorithm 2

●

Use GA via MATLAB: Case Studies

Video • 5 min

●

Materials on Case Studies

Reading • 10 min

●

Solving Real Engineering Problems with Genetic Algorithms

Reading • 5 min

Module 5 Quiz

●

Module 5 Quiz

Practice Assignment • 30 min

Module 5 Assignment

●

Module 5 Assignment

Graded App Item • 30 min

●

Objective function for Assignment 5

Reading • 5 min

●

Applying Genetic Algorithms to a Real Engineering Design Problem

Reading • 5 min

●

Designing and Applying a Genetic Algorithm to Solve an Engineering Optimization Problem

Dialogue • 5 min

●

Module 5 Recap

Reading • 5 min

Module 3

Particle Swarm Optimization

How GA Operators Work Together

 0:03 / 0:56 

2x





In this section, you studied the internal mechanisms of genetic algorithms, including solution representation, population initialization, selection, crossover, mutation, and stopping criteria.

Reflect on the following questions:

- Why does the choice of **solution representation** (binary, real-valued, permutation) strongly influence how crossover and mutation are defined?
- How does **selection** improve solution quality while simultaneously reducing diversity, and why is this trade-off unavoidable?
- Why are **crossover and mutation** both necessary in GA, and what specific problems arise if either one is missing?
- How do different stopping criteria reflect practical trade-offs between computational cost and solution quality in real engineering applications?

This discussion will help you move from viewing GA as a sequence of operators to understanding it as a coordinated balance between exploration and exploitation.

Go to next item

✔ Completed

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